

TrES-3b

R-0427

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_250709 · created 2026-07-19T09:58:48+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460865.7446 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.055
Transit depth (Rp/Rs)^2	3.68 +/- 2.11 [%]
Orbital Inclination (inc)	82.16 +/- 2.76
Airmass coefficient 1 (a1)	0.735 +/- 0.054
Airmass coefficient 2 (a2)	0.31 +/- 0.39
Scatter in the residuals of the lightcurve fit is	7.39 %
Best Comparison Star	#2 - [578, 449]
Optimal Aperture	3.22
Optimal Annulus	10.94
Transit Duration (day)	0.051 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.192 ± 0.055 vs 0.16309 (deviation 0.51σ)
Duration fit vs published	0.051 d vs 0.0567 d (deviation 0.2σ)
Mid-time O–C	-5.4 min
Depth SNR	3.4
Residual scatter	7.39 %

Light curve

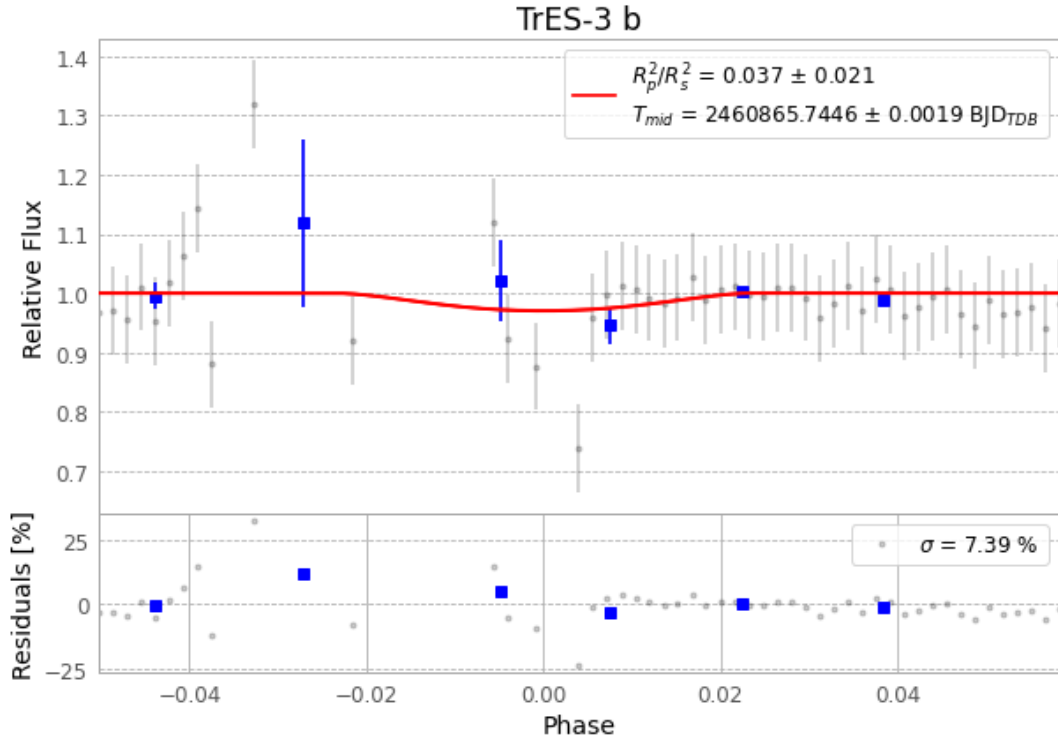


Figure 1 — FinalLightCurve_TrES-3 b_2025-07-08.png (SHA-256 2ae466eb03acf25a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [334, 233], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 b02e19f7155f958d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

69 FITS frames (45 MB), TRES-3250709041216.FITS → TRES-3250709073616.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-250709.log (3ec88eb65141...), FinalLightCurve_TrES-3 b_2025-07-08.png (2ae466eb03ac...), FinalLightCurve_TrES-3 b_2025-07-08.csv (cca91f73ed5b...)

Compute resources

Wall time 210.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 09:58Z.